

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	138781	46.072511
1	[Git, Change, Create]	44929	14.915528
2	[Git, Change, Delete]	22781	7.562836
3	[Java, ByteCode, Method, Member]	13528	4.491025
4	[Java, ByteCode, Parameter]	13306	4.417325
5	[Git, Commit]	11390	3.781252
6	[File, Git]	8286	2.750786
7	[Java, ByteCode, Bound, ParameterizedType]	7286	2.418806
8	[Java, ByteCode, Bound]	7245	2.405195
9	[Git, Change, Rename]	7018	2.329835
10	[Java, ByteCode, Member, Field]	3606	1.197120
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979341
12	[Java, Value, ByteCode, Annotation]	2936	0.974693
13	[Xml, Element]	2162	0.717741
14	[Java, ByteCode, Constructor, Method, Member]	2113	0.701474
15	[Xml, Text]	1450	0.481371
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.368830
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321357
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295462
19	[Type, File, Java, ByteCode, Class]	845	0.280523
20	[Json, Key]	739	0.245333
21	[Value, Json, Scalar]	723	0.240022
22	[Java, Value, ByteCode, Primitive]	673	0.223423
23	[Type, File, Java, ByteCode, JavaType]	649	0.215455
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191884
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136776
26	[Git, Change, Copy]	298	0.098930
27	[Author, Git, Person]	295	0.097934
28	[Value, Array]	289	0.095942
29	[Committer, Git, Person]	242	0.080339

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

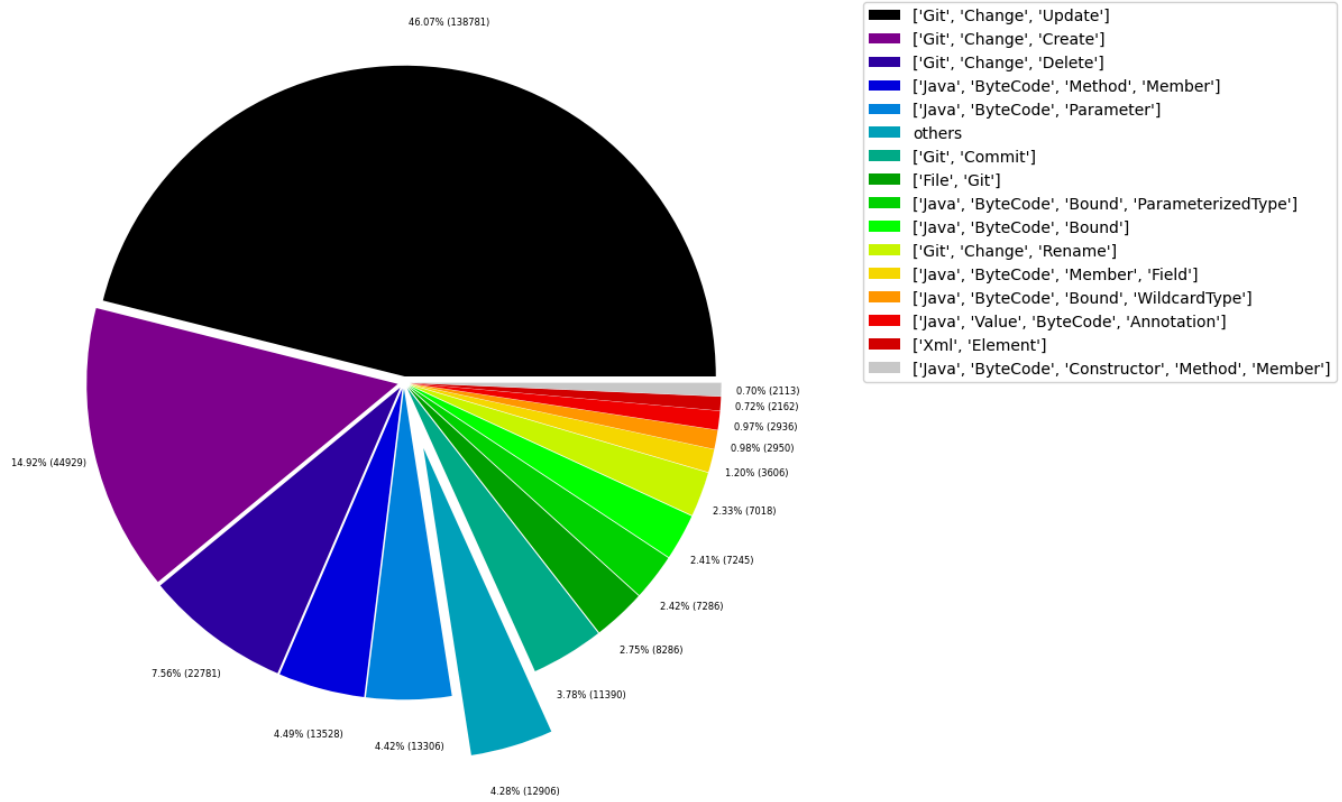


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[Type, File, Java, ByteCode, Void]	9	0.002988
9	[File, Maven, Xml, Pom, Document]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005312
16	[Maven, PluginExecution]	16	0.005312
17	[Xml, Attribute]	18	0.005976
18	[JQAssistant, Rule, Concept]	19	0.006308
19	[Maven, Plugin]	21	0.006972
20	[Maven, Configuration]	21	0.006972
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006972
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007304
23	[Type, File, Java, ByteCode, Enum]	28	0.009295
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009959
25	[Xml, Namespace]	36	0.011951
26	[Type, File, Java, ByteCode, Annotation]	44	0.014607
27	[File, Directory]	44	0.014607
28	[Type, File, Java, ByteCode, Class, Throwable]	55	0.018259
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.028218

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

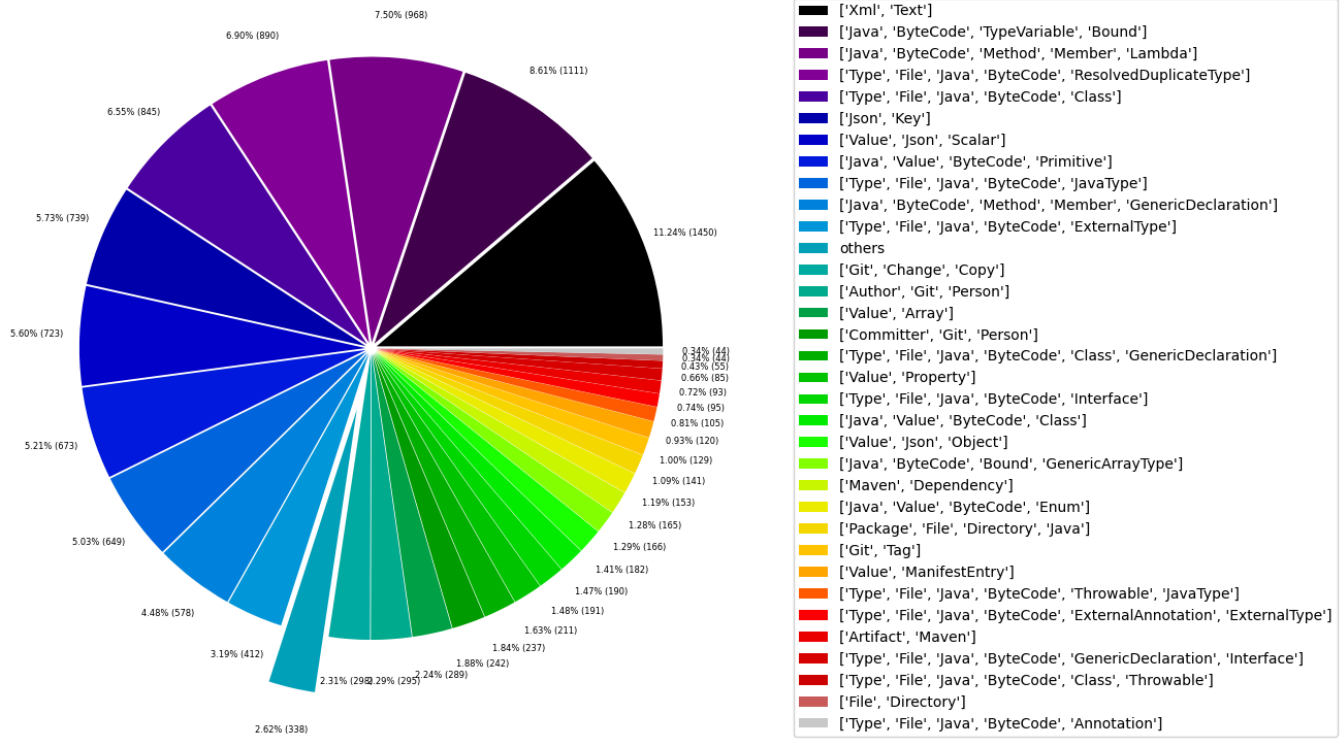


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

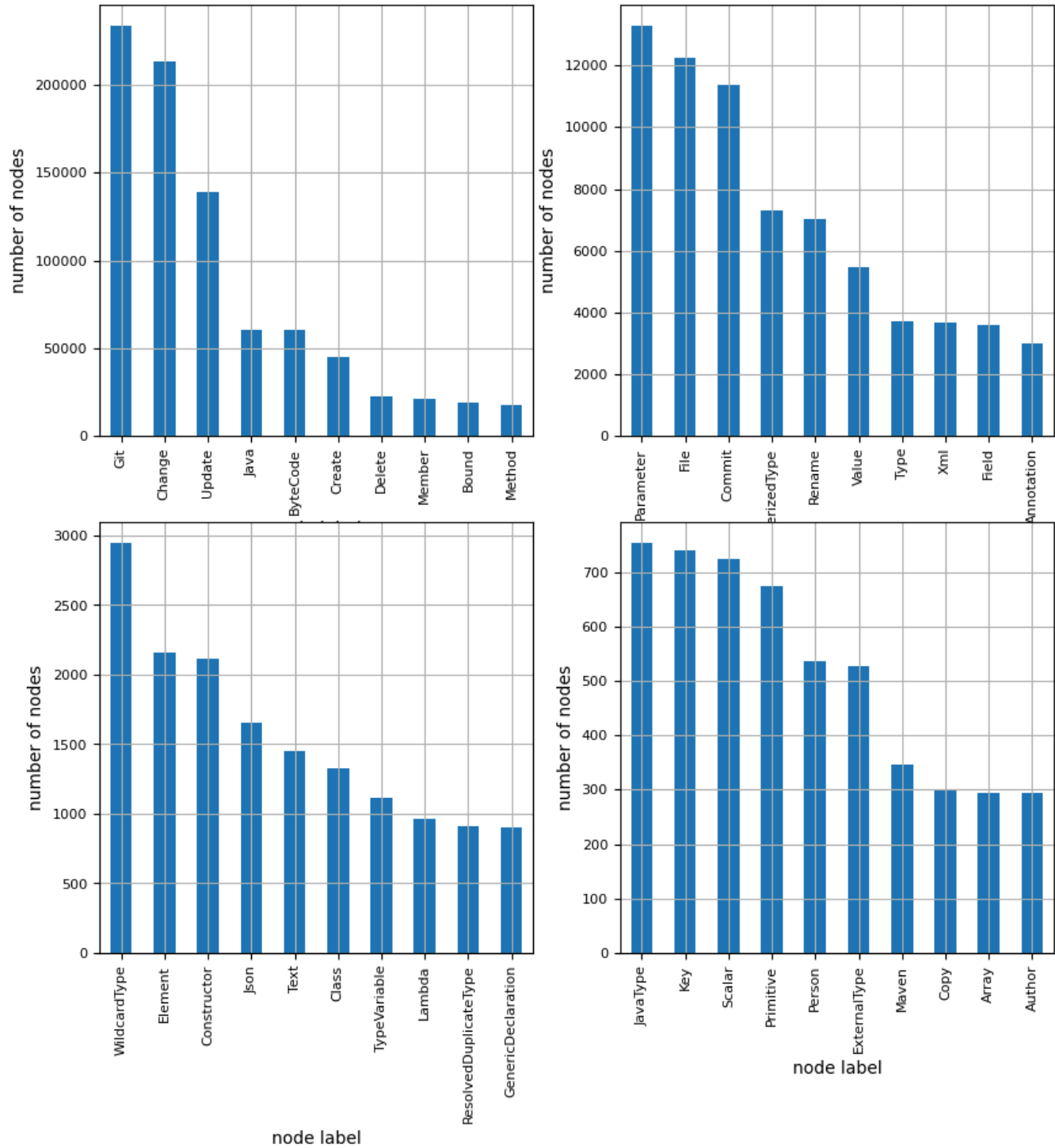
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.733440
1	Change	213807	70.979640
2	Update	138781	46.072511
3	Java	60721	20.158155
4	ByteCode	60531	20.095079
5	Create	44929	14.915528
6	Delete	22781	7.562836
7	Member	20797	6.904187
8	Bound	18758	6.227280
9	Method	17191	5.707068
10	Parameter	13306	4.417325
11	File	12244	4.064763
12	Commit	11390	3.781252
13	ParameterizedType	7286	2.418806
14	Rename	7018	2.329835
15	Value	5483	1.820246
16	Type	3718	1.234301
17	Xml	3675	1.220026
18	Field	3606	1.197120
19	Annotation	2980	0.989300
20	WildcardType	2950	0.979341
21	Element	2162	0.717741
22	Constructor	2117	0.702802
23	Json	1652	0.548431
24	Text	1450	0.481371
25	Class	1327	0.440537
26	TypeVariable	1111	0.368830
27	Lambda	968	0.321357
28	ResolvedDuplicateType	912	0.302766
29	GenericDeclaration	904	0.300110
30	JavaType	754	0.250313
31	Key	739	0.245333
32	Scalar	723	0.240022
33	Primitive	673	0.223423
34	Person	537	0.178273
35	ExternalType	528	0.175285
36	Maven	346	0.114865
37	Copy	298	0.098930
38	Array	295	0.097934
39	Author	295	0.097934

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 936048

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.841457
1	MODIFIES	213807	22.841457
2	UPDATES	138781	14.826270
3	CREATES	52245	5.581445
4	INVOKES	36694	3.920098
5	DELETES	29799	3.183491
6	COMMITTED	22780	2.433636
7	DEPENDS_ON	22504	2.404150
8	OF_TYPE	21945	2.344431
9	DECLARES	21257	2.270931
10	OF_RAW_TYPE	17367	1.855354
11	HAS	14444	1.543083
12	HAS_PARENT	13921	1.487210
13	RETURNS	12858	1.373648
14	HAS_COMMIT	11390	1.216818
15	READS	9385	1.002620
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898886
17	HAS_FILE	8286	0.885211
18	RENAMES	7018	0.749748
19	OF_GENERIC_TYPE	6000	0.640993
20	RESOLVES_TO	4247	0.453716
21	SIMILAR	4078	0.435661
22	WRITES	3948	0.421773
23	CONTAINS	3928	0.419637
24	HAS_NEW_NAME	3662	0.391219
25	RETURNS_GENERIC	3597	0.384275
26	CHANGED_TOGETHER_WITH	3034	0.324129
27	ANNOTATED_BY	2924	0.312377
28	REQUIRES	2233	0.238556
29	HAS_FIRST_CHILD	2162	0.230971

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

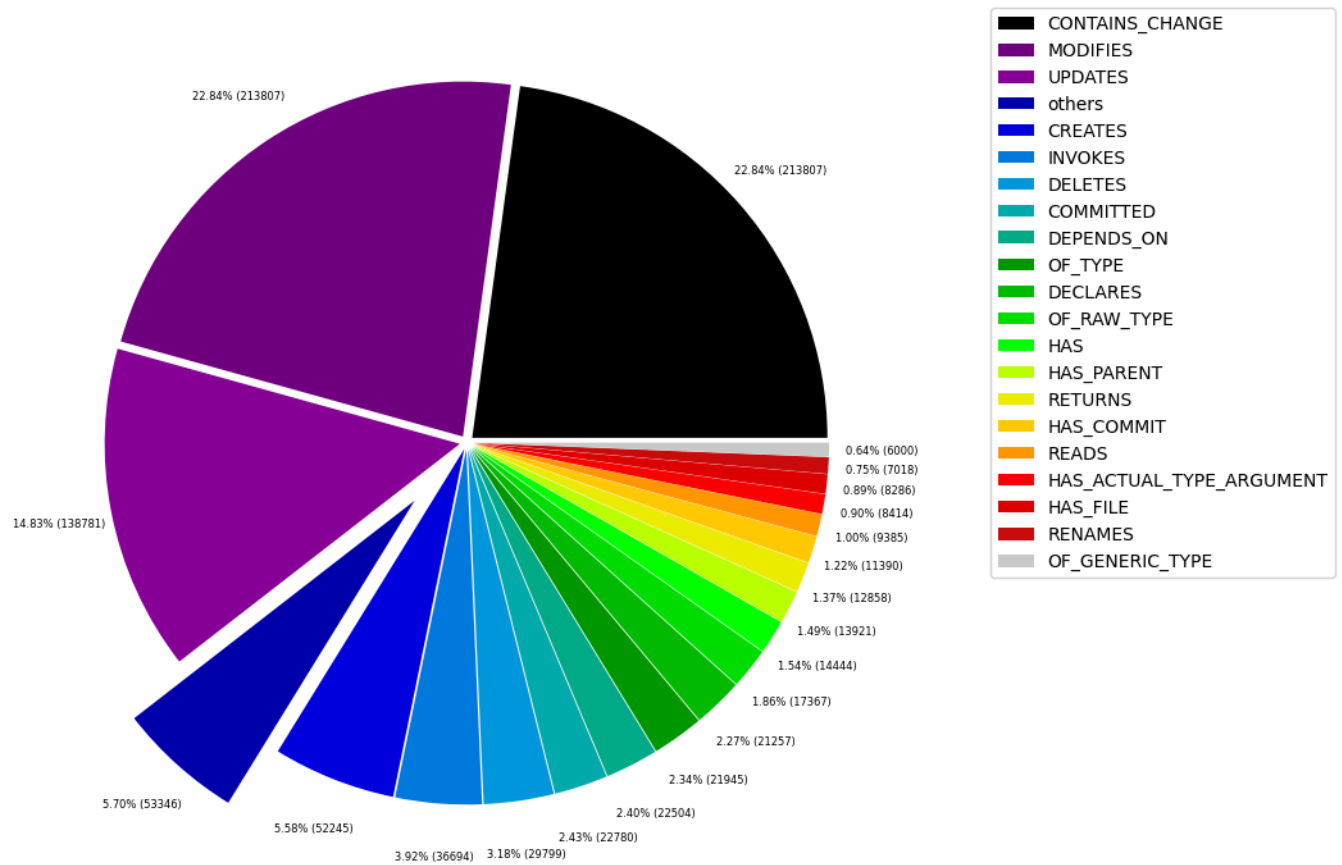


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

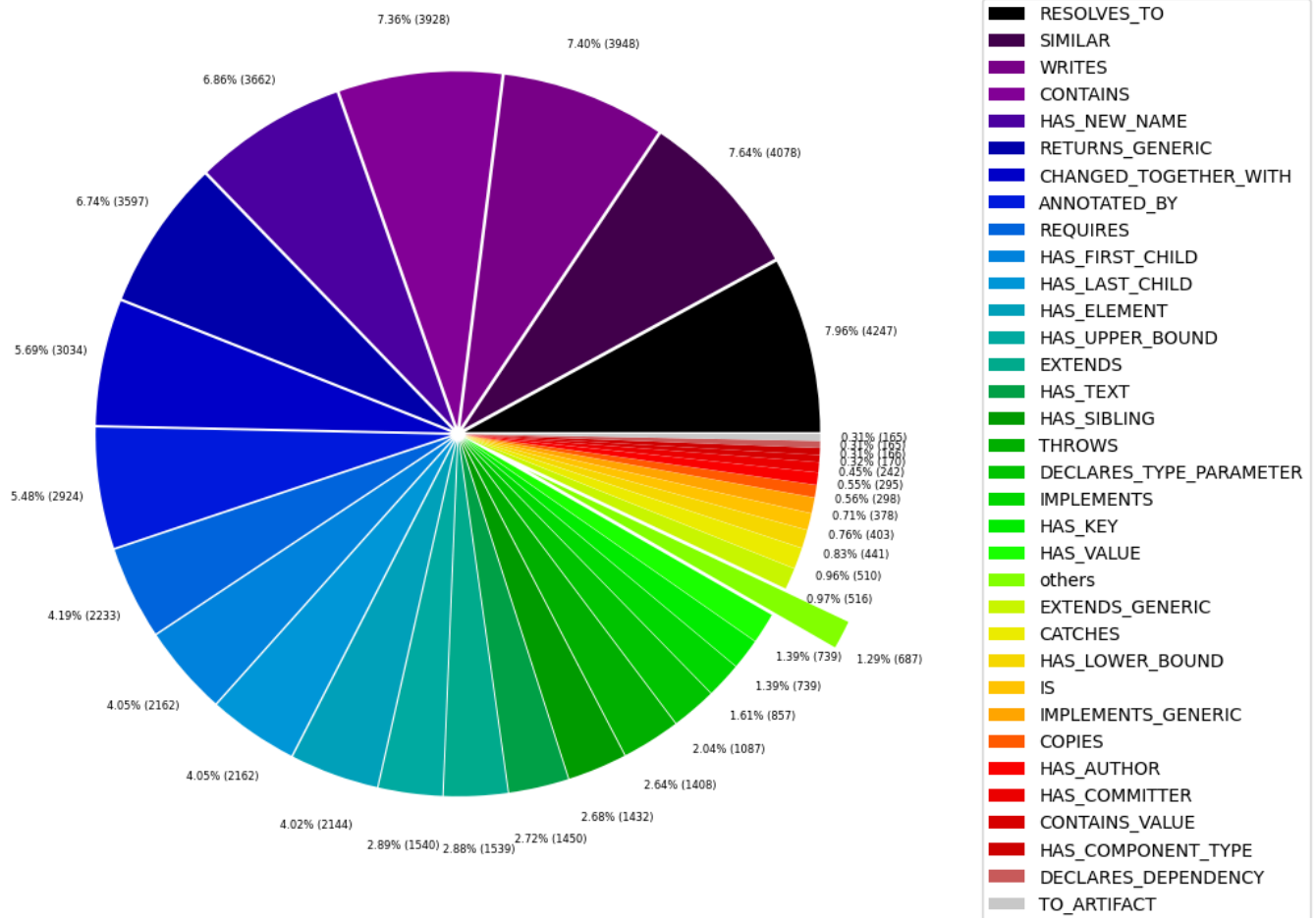
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000961
6	HAS_ROOT_ELEMENT	10	0.001068
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002243
13	IS_ARTIFACT	21	0.002243
14	HAS_CONFIGURATION	21	0.002243
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002991
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004166
19	COPY_OF	119	0.012713
20	HAS_TAG	129	0.013781
21	ON_COMMIT	129	0.013781
22	DECLARES_DEPENDENCY	165	0.017627
23	TO_ARTIFACT	165	0.017627
24	HAS_COMPONENT_TYPE	166	0.017734
25	CONTAINS_VALUE	170	0.018161
26	HAS_COMMITTER	242	0.025853
27	HAS_AUTHOR	295	0.031515
28	COPIES	298	0.031836
29	IMPLEMENTS_GENERIC	378	0.040383

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	138781	11390	
1	[Git, Change, Update]	MODIFIES	[File, Git]	138781	138781	
2	[Git, Change, Update]	UPDATES	[File, Git]	138781	138781	
3	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
4	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22339	13429	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8516	13429	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8430	13429	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3466	7245	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3114	7286	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2113	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301223
total_number_of_relationships (edges): 936048
-> total directed graph density: 1.0316284307903962e-05
-> total directed graph density in percent: 0.0010316284307903961
```